

① Line

2) Polyline

3) Circle

4) Arc

5) Rectangle

6) Ellipse

7) Hatch

8) Move

9) Copy

10) Stretch

11) Rotate

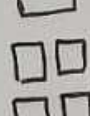
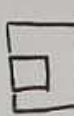
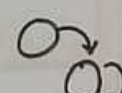
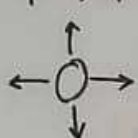
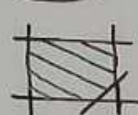
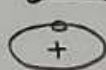
12) Mirror

13) Scale

14) Trim

15) Fillet

16) Array



## Auto Cad Symbols:-

17) Erase



18) Explode



19) Offset



Sketch free hand the graphical symbols for the following Engineering items.

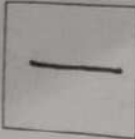
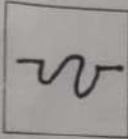
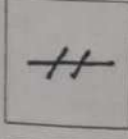
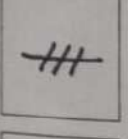
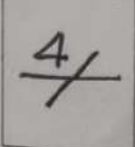
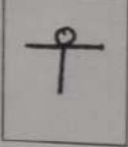
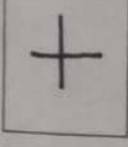
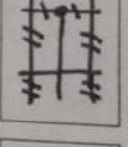
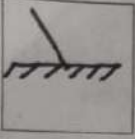
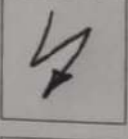
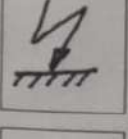
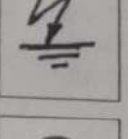
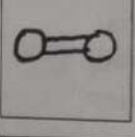
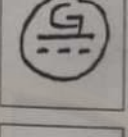
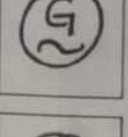
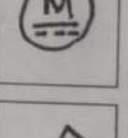
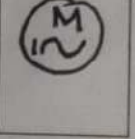
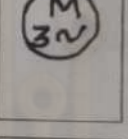
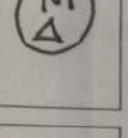
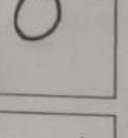
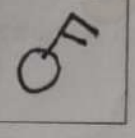
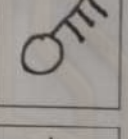
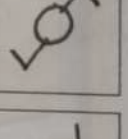
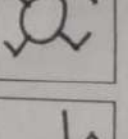
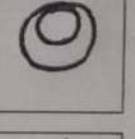
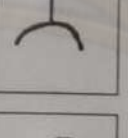
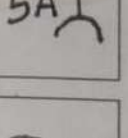
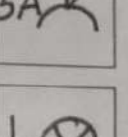
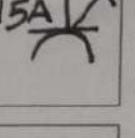
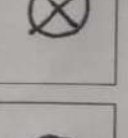
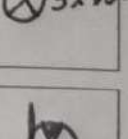
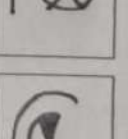
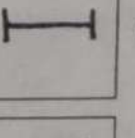
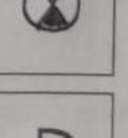
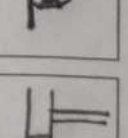
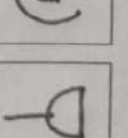
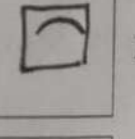
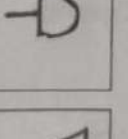
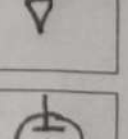
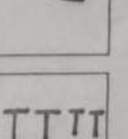
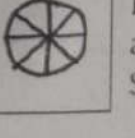
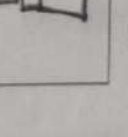
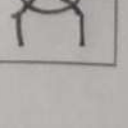
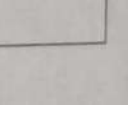
## Electronics and Electrical

|                                           |  |                     |  |                        |  |                         |  |
|-------------------------------------------|--|---------------------|--|------------------------|--|-------------------------|--|
| Amplifier                                 |  | Antenna             |  | Arrester               |  | Battery                 |  |
| Circuit Breaker                           |  | Capacitor           |  | Coil                   |  | Connector               |  |
| Electric Contact                          |  | Core                |  | Fuse                   |  | Directional Coupler     |  |
| Ground                                    |  | Handset             |  | Rectifier              |  | Visual Signaling device |  |
| Receiver                                  |  | Repeater            |  | Ground                 |  | Handset                 |  |
| Thermocouple                              |  | Inductor Winding    |  | Incandescent Lamp      |  | Ballast Lamp            |  |
| Transformer                               |  | Switch              |  | Resistor               |  | Transmission Path       |  |
| Generator                                 |  | Motor               |  | Solenoid               |  | Thermostat              |  |
| Thermistor                                |  | Variable Resistor   |  | Voltmeter              |  | NPN-type Transistor     |  |
| PNP-type Transistor                       |  | Direct Current      |  | Alternating Current    |  | Power Frequency         |  |
| Apparatus & Machine suitable for DC or AC |  | Half-wave Rectifier |  | Line or Cable existing |  | Line or Cable planned   |  |
| Silicon Controlled Rectifier              |  | Power Line          |  | Underground Cable      |  | Overhead Line           |  |

# Electronics and Electrical

|                                     |  |                          |  |                                        |  |                                     |  |
|-------------------------------------|--|--------------------------|--|----------------------------------------|--|-------------------------------------|--|
| Conductors or a group of Conductors |  | Flexible Conductors      |  | Two Conductors                         |  | Three Conductors                    |  |
| Four Conductors                     |  | Junction of Conductors   |  | Crossing without Electrical Connection |  | Crossing and Connecting Conductors  |  |
| Frame and Chassis Connection        |  | Fault                    |  | Fault to Frame                         |  | Earth Fault                         |  |
| Mechanically Coupled Machine        |  | DC Generator             |  | AC Generator                           |  | DC Motor                            |  |
| AC Motor Single Phase               |  | Three Phase Motor        |  | Three Phase Motor in Delta Connection  |  | One Way Switch (Single Pole Switch) |  |
| Two Pole Switch                     |  | Three Pole Switch        |  | Two Way Switch                         |  | Intermediate Switch                 |  |
| Push Button Switch                  |  | Socket Outlets           |  | Socket Outlets, 5A                     |  | Socket Outlets, 5A with Switch      |  |
| Socket Outlets, 15A with Switch     |  | Lamp mounted on ceiling  |  | Group of 3, 40 Watt Lamp               |  | Lamp mounted on wall                |  |
| Fluorescent Lamp                    |  | Ceiling Fan              |  | Wall mounted Fan                       |  | Exhaust Fan                         |  |
| Fan Regulator                       |  | Bell                     |  | Pickup                                 |  | Buzzer                              |  |
| Siren                               |  | Public addressing System |  | Diode with Filament                    |  | Telephone-Telegraph Line            |  |

# Electronics and Electrical

|                                     |                                                                                     |                          |                                                                                     |                                        |                                                                                       |                                     |                                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------|
| Conductors or a group of Conductors |    | Flexible Conductors      |    | Two Conductors                         |    | Three Conductors                    |    |
| Four Conductors                     |    | Junction of Conductors   |    | Crossing without Electrical Connection |    | Crossing and Connecting Conductors  |    |
| Frame and Chassis Connection        |    | Fault                    |    | Fault to Frame                         |    | Earth Fault                         |    |
| Mechanically Coupled Machine        |    | DC Generator             |    | AC Generator                           |    | DC Motor                            |   |
| AC Motor Single Phase               |   | Three Phase Motor        |   | Three Phase Motor in Delta Connection  |  | One Way Switch (Single Pole Switch) |  |
| Two Pole Switch                     |  | Three Pole Switch        |  | Two Way Switch                         |  | Intermediate Switch                 |  |
| Push Button Switch                  |  | Socket Outlets           |  | Socket Outlets, 5A                     |  | Socket Outlets, 5A with Switch      |  |
| Socket Outlets, 15A with Switch     |  | Lamp mounted on ceiling  |  | Group of 3, 40 Watt Lamp               |  | Lamp mounted on wall                |  |
| Fluorescent Lamp                    |  | Ceiling Fan              |  | Wall mounted Fan                       |  | Exhaust Fan                         |  |
| Fan Regulator                       |  | Bell                     |  | Pickup                                 |  | Buzzer                              |  |
| Siren                               |  | Public addressing System |  | Diode with Filament                    |  | Telephone-Telegraph Line            |  |

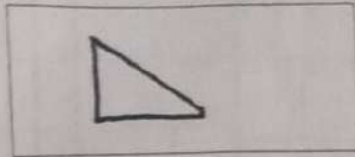
the graphical symbols for the following Engineering items.

## Electronics and Electrical

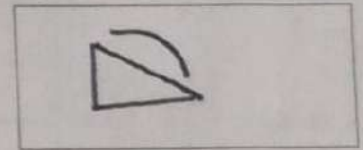
|                                           |  |                     |  |                        |  |                         |  |
|-------------------------------------------|--|---------------------|--|------------------------|--|-------------------------|--|
| Amplifier                                 |  | Antenna             |  | Arrester               |  | Battery                 |  |
| Circuit Breaker                           |  | Capacitor           |  | Coil                   |  | Connector               |  |
| Electric Contact                          |  | Core                |  | Fuse                   |  | Directional Coupler     |  |
| Ground                                    |  | Handset             |  | Rectifier              |  | Visual Signaling device |  |
| Receiver                                  |  | Repeater            |  | Ground                 |  | Handset                 |  |
| Thermocouple                              |  | Inductor Winding    |  | Incandescent Lamp      |  | Ballast Lamp            |  |
| Transformer                               |  | Switch              |  | Resistor               |  | Transmission Path       |  |
| Generator                                 |  | Motor               |  | Solenoid               |  | Thermostat              |  |
| Thermistor                                |  | Variable Resistor   |  | Voltmeter              |  | NPN-type Transistor     |  |
| PNP-type Transistor                       |  | Direct Current      |  | Alternating Current    |  | Power Frequency         |  |
| Apparatus & Machine suitable for DC or AC |  | Half-wave Rectifier |  | Line or Cable existing |  | Line or Cable planned   |  |
| Controlled Rectifier                      |  | Power Line          |  | Underground Cable      |  | Overhead Line           |  |

Sketch freehand the graphical symbols for the following welding items.

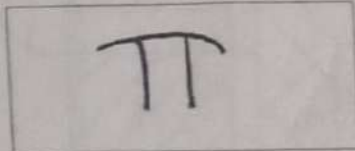
Lap Weld



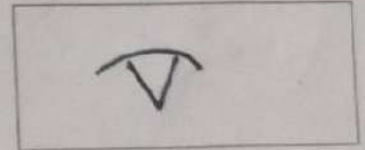
Fillet



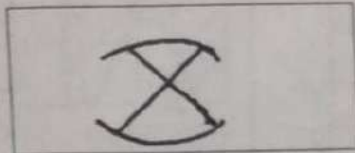
Square Butt



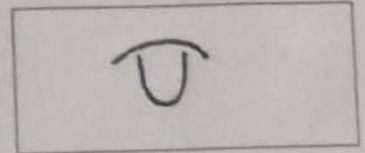
Single V-Butt



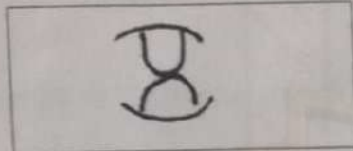
Double V-Butt



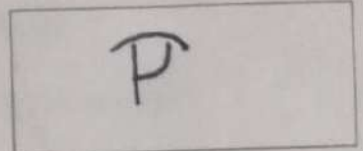
Single U-Butt



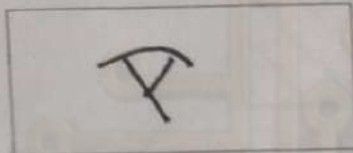
Double U- Butt



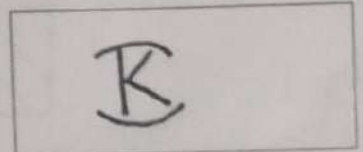
Single J-Butt



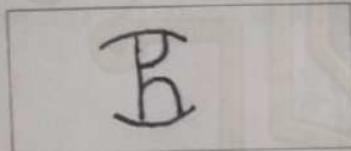
Single Bevel Butt



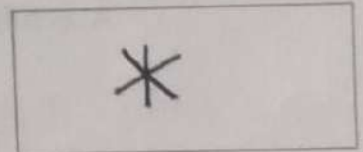
Double Bevel Butt



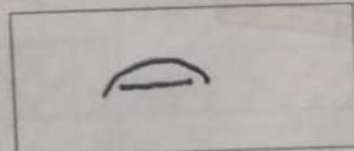
Double J-Butt



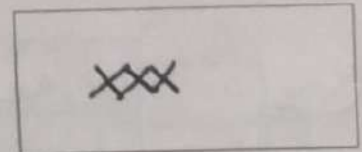
Spot Weld



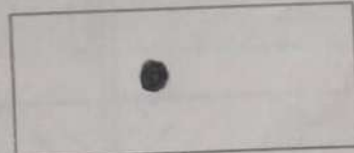
Bead or Edge Weld



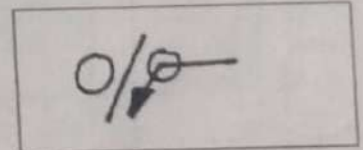
Seam Weld



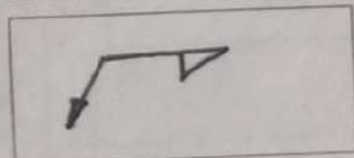
Field Weld



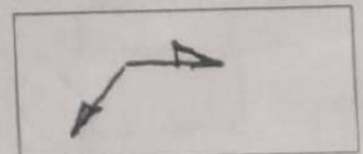
Weld all around



Fillet Weld on own side of joints



Fillet weld on opposite side of joint



Fillet weld on both sides of joint

